

Types of solid materials

Composites:

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Composites are two or more chemically distinct materials which are combined together have improved properties over the individual materials.

properties of composite materials

1. High specific strength
2. High specific stiffness
3. Low density
4. Corrosion resistance
5. Wear resistance
6. Fatigue resistance
7. Creep resistance
8. Light in weight
9. High ductility
10. ^{low} Good thermal conductivity

12. Easier to produce 13. Easy bondable

Applications of composites materials

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1. Fiberglass is used in marine body, plastic pipes, storage containers and industrial floorings.
2. Carbon reinforced plastic are utilized extensively in sports and recreational equipment, pressure vessels and aircraft structural components
3. Silicon carbide and alumina are used in tennis, rockets and circuit boards.
4. concrete is used as structural building material.

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Application of metals

1. Platinum is used in chemical laboratory to measure elevated temperature
2. Steels are used in knives, blades, springs, cutters, drills etc.

3. Aluminium are used in aircraft structures, fly wheel, truck, marine structures and storage equipment.

4. Copper metals are utilized in electrical wire, fuse chips, welding rods etc.

5. Magnesium metals are used in many devices like chain saws, power tools, clippers etc.

Applications of ceramics

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1. Glass is used in containers, windows, lenses, and fibreglass dinnerware.

2. Ceramics materials are used in spacecraft. The ceramics tiles made from 99.7% pure silica fibers are used for the space shuttle.

3. Silicon carbide is utilized in the high temperature areas of the experimental AGT-100 automotive gas and turbine engine.

4. Aluminium oxide is used in the support base for integrated circuit chip in a thermal conduction.

5. Tradition ceramics bricks and tiles are used in the construction industries and electrical porcelain in the electrical industry.

6. Abrasive ceramics (diamond, sic, wc, Al_2O_3) are used to wear grind or cut away other material which necessarily is softer.

Applications of polymers

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1. Polysulfones, polystyrenes are used as electrical connectors such as switches, TV tuner components, high voltage components, terminal boards, integrated circuits boards, capacitors etc.
2. Nylon are used in almost all industries include gears, bearings and antifriction parts.
3. Polymeric materials are utilized in automobiles such as fuel systems, seat belts, window handles etc.
4. Polymeric materials ^{are} also used in photographic film, flexible and sterilizable bottles, luggage, magnetic recording tapes, telephones, battery parts, medical instruments etc.

Semiconductor

A semiconductor is a material which has a conductivity between conductors and insulators. Semiconductors can be pure materials/elements, such as Si, Ge or compounds such as gallium arsenide (GaAs) and cadmium selenide (CdSe).

Characteristics of semiconductor

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1. Conductivity is between conductor and insulator.
2. Band gap energy is medium. ($0.2 \text{ eV} < E_g < 3.0 \text{ eV}$)
3. Fourth group ^(IVA) elements of periodic table.
4. Has covalent bond
5. Stable state is octate form.
6. ~~Full filled valence~~ Empty conduction band
7. Negative co-efficient of resistivity.
8. At absolute zero temperature, all semiconductors are insulators.

Applications of semi conductor

1. Transistors
2. Solar cells
3. Diodes
4. Digital devices
5. Automobiles
6. Airtel Aircraft

Super conductivity

Super conductivity is the property of certain materials to conduct direct current (DC) electricity without energy loss when they are cooled below a critical temperature. The substances that have super conductivity are called super conductors. At -269°C , the resistance of Hg and at -266°C , the resistance of Sn are zero.

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crystalline solid

A crystalline solid is one in which the atoms are situated in a repeating or periodic array over large atomic distances.

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Amorphous / non-crystalline solid

Amorphous is a non-crystalline solid in which the atoms and molecules are not organized in a definite lattice pattern.

crystalline Solids	Amorphous Solids
1. Long range orderly arrangement of constituents.	1. Short range, random arrangement of constituents.
2. Definite / regular shape	2. Irregular shape
3. anisotropic in nature	3. isotropic in nature.
4. Definite value of heat of fusion.	4. doesn't have definite value of heat of fusion.
5. They have sharp melting points.	5. Gradually soften over a range of temperature and so can be moulded.
Ex - NaCl, diamond	Ex - Rubber, plastics, glass etc.

Applications of super conductor

1. Medical MRI/NMR devices
2. Magnetic energy storage system.
3. Motors
4. Generators
5. Transformers
6. Computer parts
7. Sensitive devices for the measurement of magnetic fields
8. Electrical currents etc.

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Properties of super conductor

1. has zero electrical resistance
2. has effect of pressure
3. has effect of impurity
4. Brittle in nature
5. exhibit perfect diamagnetism
6. No energy loss
7. No heating loss.

The Meissner effect in superconductor

The Meissner effect is the expulsion of a magnetic field from a superconductor during its transition to the superconducting state when it is cooled below the critical temperature. This expulsion will repel a nearby magnet. The Meissner effect equation is

$$B = \mu_0 (H + I) \quad \text{where,}$$

$$\therefore B = 0$$

$I = \text{Magnetization}$

$H = \text{external applied magnetic field}$

Advance Materials

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Materials utilized in high technology applications are sometimes termed as advance materials.

This advance materials are typically traditional materials whose properties have been enhanced and also newly developed, high performance materials. Furthermore, they may be of all material types and are normally expensive.

Ex - Semiconductors, Biomaterials, Smart materials, Nano materials.